Schema to be Created

```
CREATE TABLE Customers (
        CustomerID NUMBER PRIMARY KEY,
        Name VARCHAR2(100),
        DOB DATE,
        Balance NUMBER,
        LastModified DATE
);

CREATE TABLE Accounts (
        AccountID NUMBER PRIMARY KEY,
        CustomerID NUMBER,
        AccountType VARCHAR2(20),
        Balance NUMBER,
        LastModified DATE,
        FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

CREATE TABLE Transactions (
        TransactionID NUMBER PRIMARY KEY,
        AccountID NUMBER,
        TransactionDate DATE,
        Amount NUMBER,
        TransactionType VARCHAR2(10),
        FOREIGN KEY (AccountID) REFERENCES Accounts(AccountID)
);

 CREATE TABLE Loans (
        LoanID NUMBER PRIMARY KEY,
        CustomerID NUMBER,
        LoanAmount NUMBER,
        InterestRate NUMBER,
        StartDate DATE,
        EndDate DATE,
        FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

CREATE TABLE Employees (
        EmployeeID NUMBER PRIMARY KEY,
        Name VARCHAR2(100),
        Position VARCHAR2(50),
        Salary NUMBER,
```

```
    Department VARCHAR2(50),
    HireDate DATE
);
```

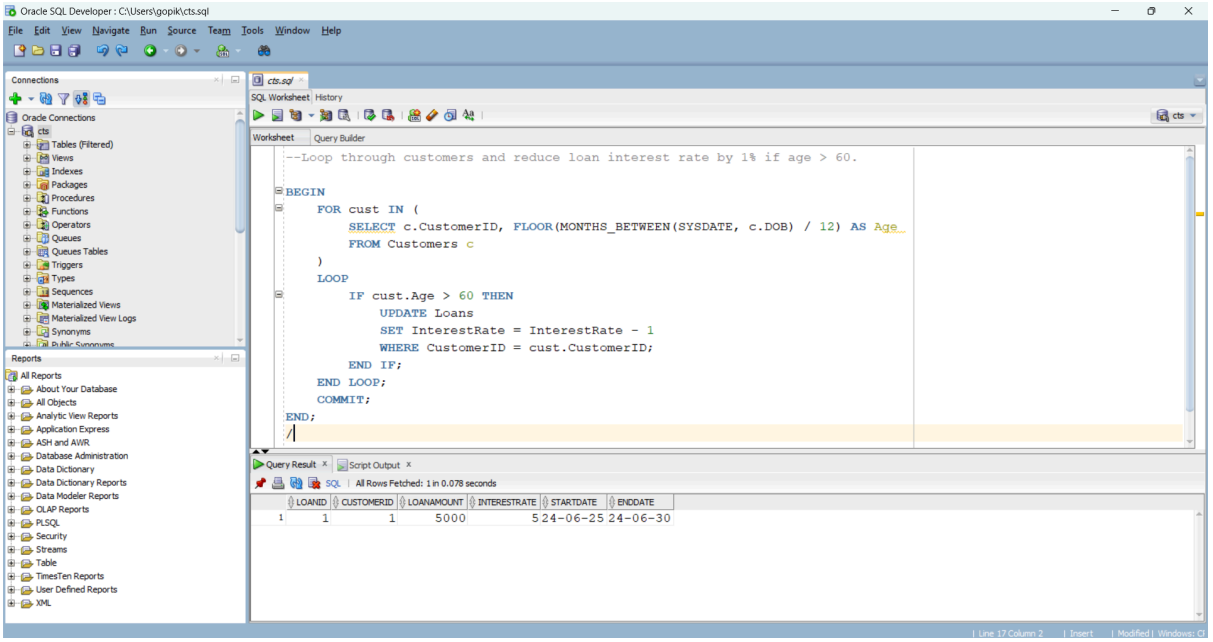
Example Scripts for Sample Data Insertion

```
INSERT INTO Customers (CustomerID, Name, DOB, Balance, LastModified)
VALUES (1, 'John Doe', TO_DATE('1985-05-15', 'YYYY-MM-DD'), 1000, SYSDATE);

INSERT INTO Customers (CustomerID, Name, DOB, Balance, LastModified)
VALUES (2, 'Jane Smith', TO_DATE('1990-07-20', 'YYYY-MM-DD'), 1500, SYSDATE);

INSERT INTO Accounts (AccountID, CustomerID, AccountType, Balance, LastModified)
VALUES (1, 1, 'Savings', 1000, SYSDATE);

INSERT INTO Accounts (AccountID, CustomerID, AccountType, Balance, LastModified)
VALUES (2, 2, 'Checking', 1500, SYSDATE);

INSERT INTO Transactions (TransactionID, AccountID, TransactionDate, Amount,
TransactionType)
VALUES (1, 1, SYSDATE, 200, 'Deposit');

INSERT INTO Transactions (TransactionID, AccountID, TransactionDate, Amount,
TransactionType)
VALUES (2, 2, SYSDATE, 300, 'Withdrawal');

INSERT INTO Loans (LoanID, CustomerID, LoanAmount, InterestRate, StartDate, EndDate)
VALUES (1, 1, 5000, 5, SYSDATE, ADD_MONTHS(SYSDATE, 60));

INSERT INTO Employees (EmployeeID, Name, Position, Salary, Department, HireDate)
VALUES (1, 'Alice Johnson', 'Manager', 70000, 'HR', TO_DATE('2015-06-15',
'YYYY-MM-DD'));

INSERT INTO Employees (EmployeeID, Name, Position, Salary, Department, HireDate)
VALUES (2, 'Bob Brown', 'Developer', 60000, 'IT', TO_DATE('2017-03-20',
'YYYY-MM-DD'));
```

Exercise 1: Control Structures

Scenario 1: The bank wants to apply a discount to loan interest rates for customers above 60 years old.

- **Question:** Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.



Scenario 2: A customer can be promoted to VIP status based on their balance.

- **Question:** Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over $10,000.

Scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.

- **Question:** Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.